
EEM 561 Machine Vision

Lecture 8

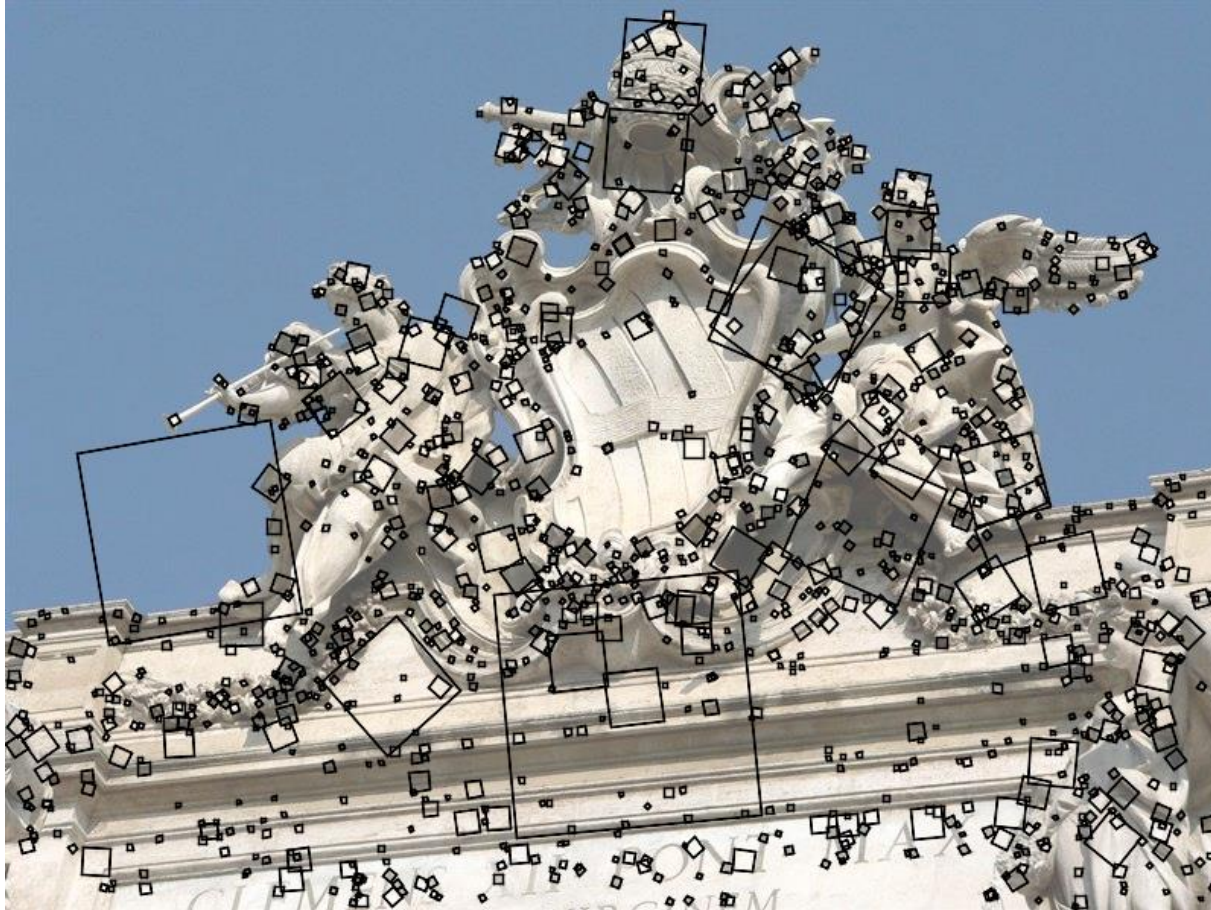
SIFT - keypoints

Spring 2025

Asst. Prof. Dr. Onur Kılınç

Eskisehir Tech. Uni., Dept. EEE

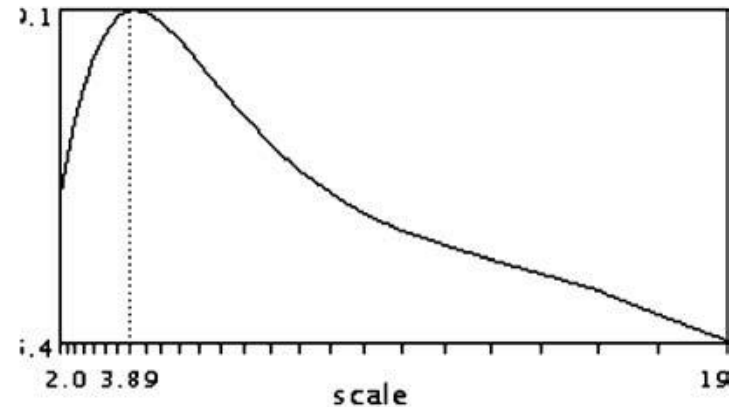
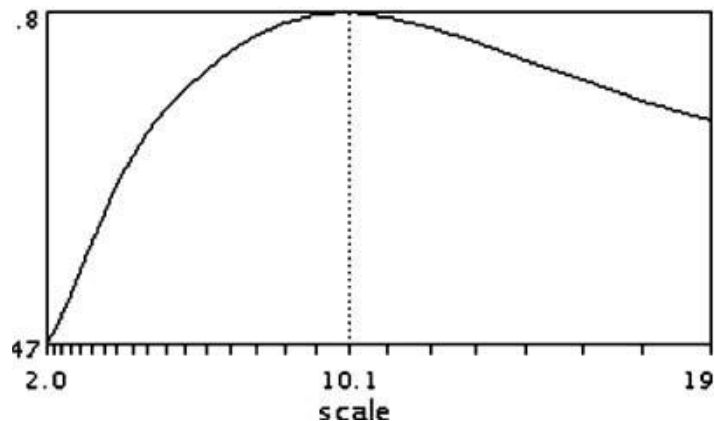
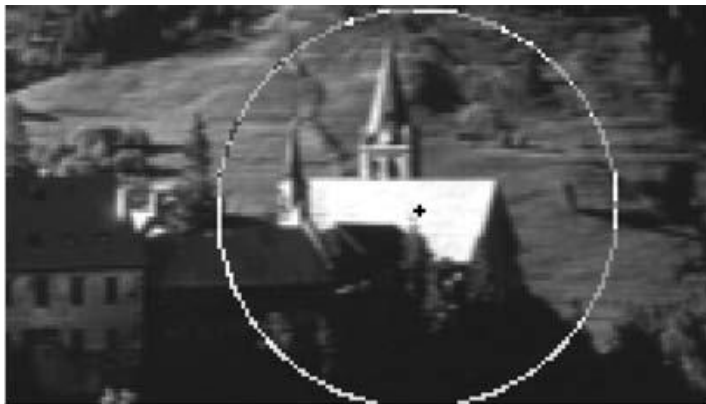
SIFT keypoint detection



D. Lowe, [Distinctive image features from scale-invariant keypoints](#),
IJCV 60 (2), pp. 91-110, 2004.

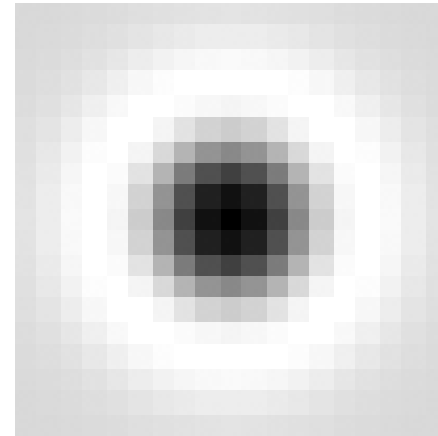
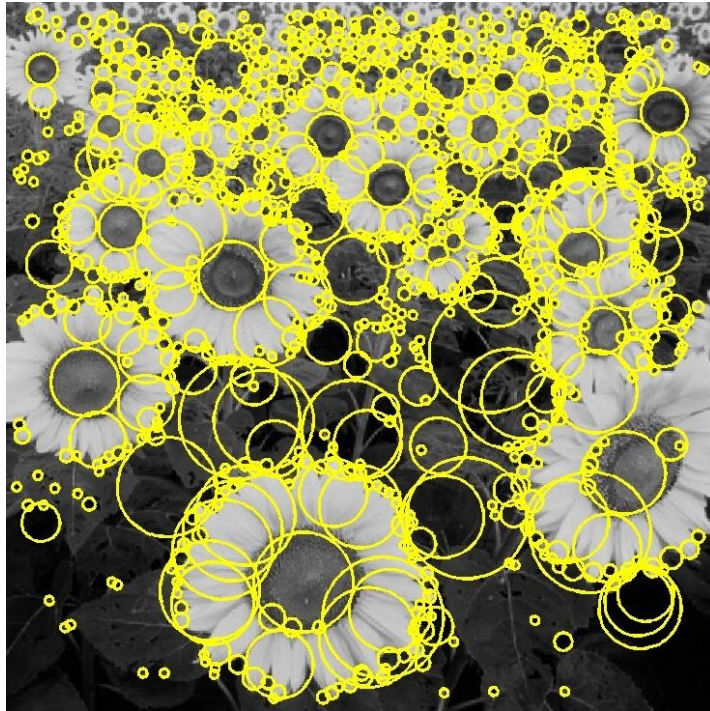
Keypoint detection with scale selection

- We want to extract keypoints with *characteristic scales* that are *covariant* w.r.t. the image transformation



Basic idea

- Convolve the image with a “blob filter” at multiple scales and look for extrema of filter response in the resulting *scale space*



T. Lindeberg. [Feature detection with automatic scale selection.](#)
IJCV 30(2), pp 77-116, 1998.

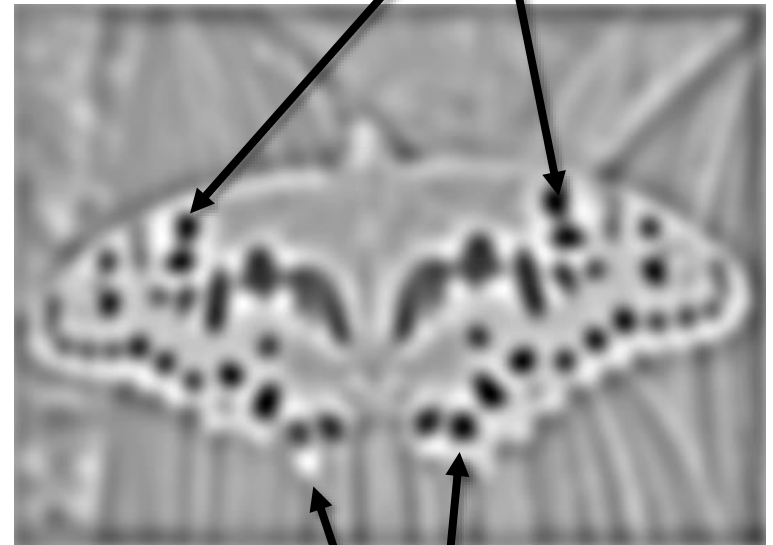
Blob detection



*



=



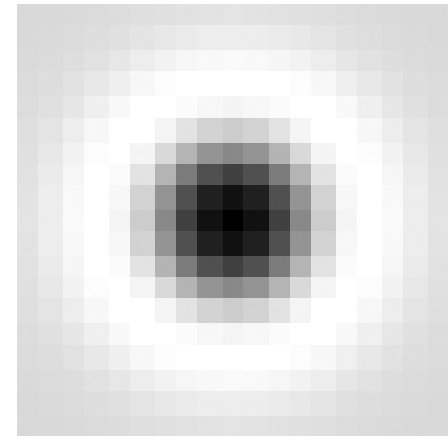
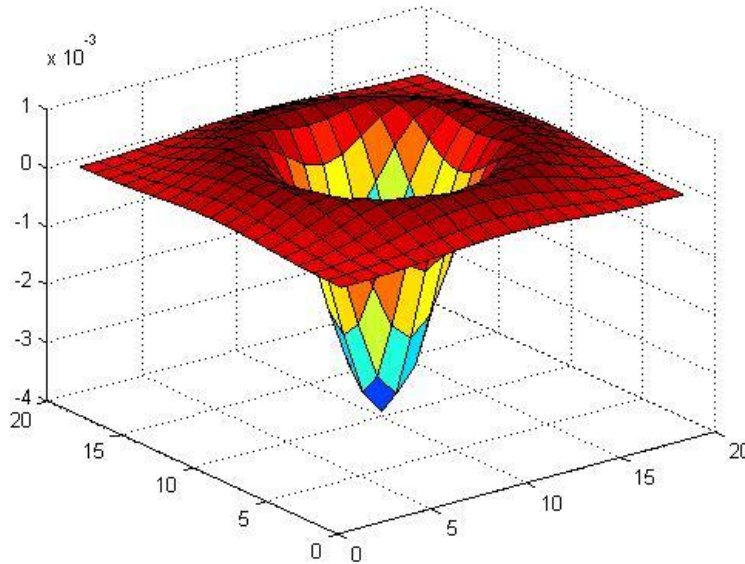
minima

maxima

Find maxima *and minima* of blob filter response
in space *and scale*

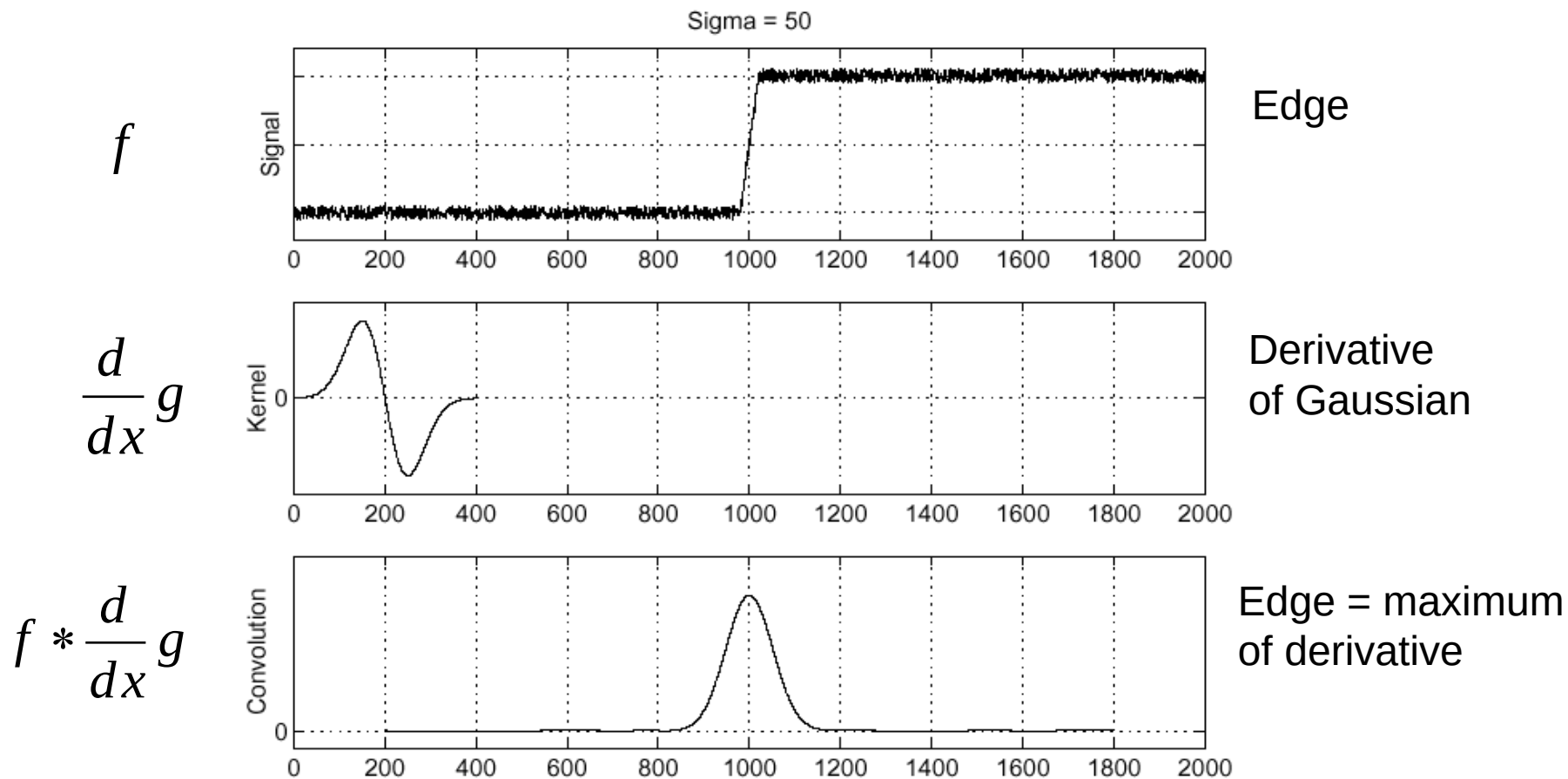
Blob filter

Laplacian of Gaussian: Circularly symmetric operator for blob detection in 2D

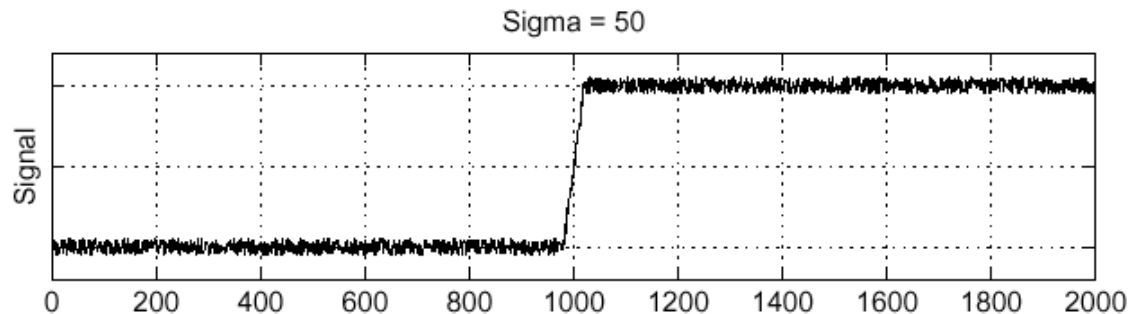


$$\nabla^2 g = \frac{\partial^2 g}{\partial x^2} + \frac{\partial^2 g}{\partial y^2}$$

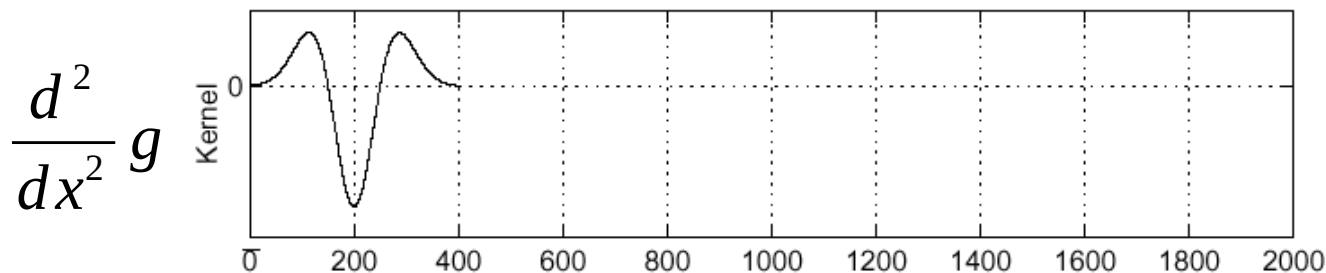
Recall: Edge detection



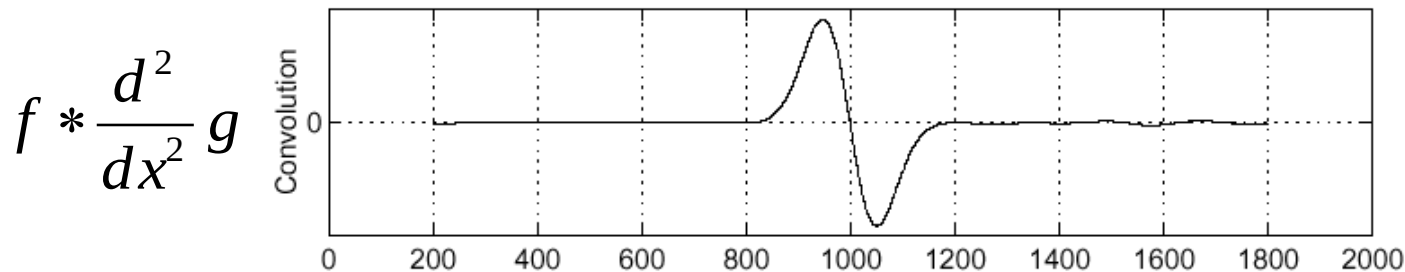
Edge detection, Take 2



Edge



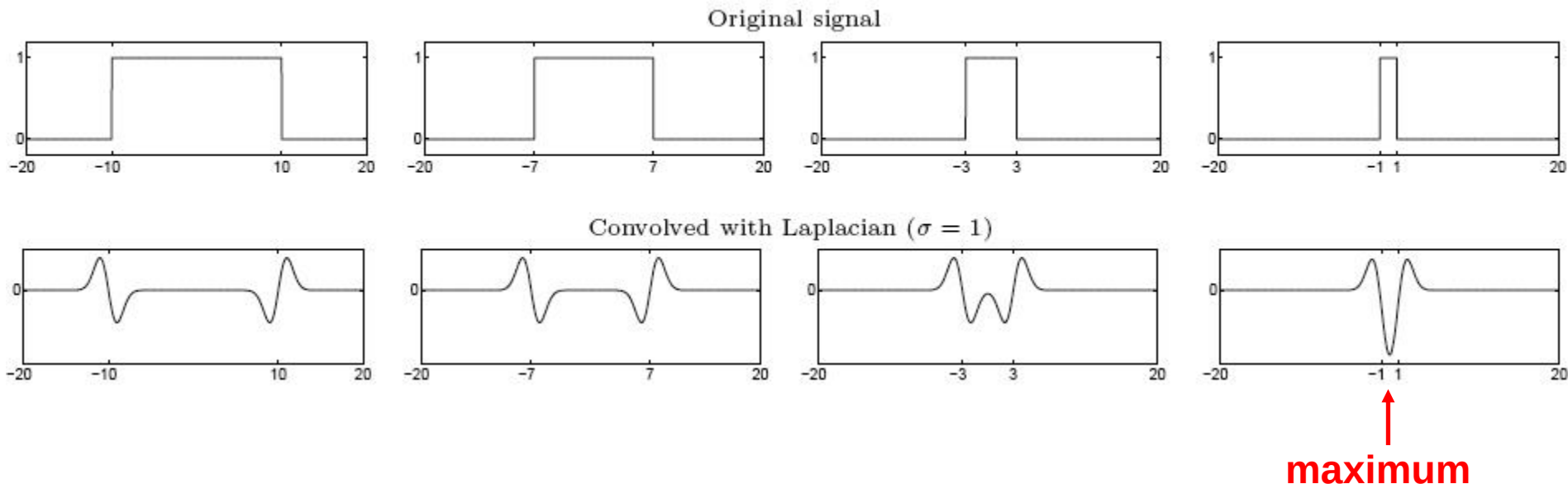
Second derivative
of Gaussian
(Laplacian)



Edge = zero crossing
of second derivative

From edges to blobs

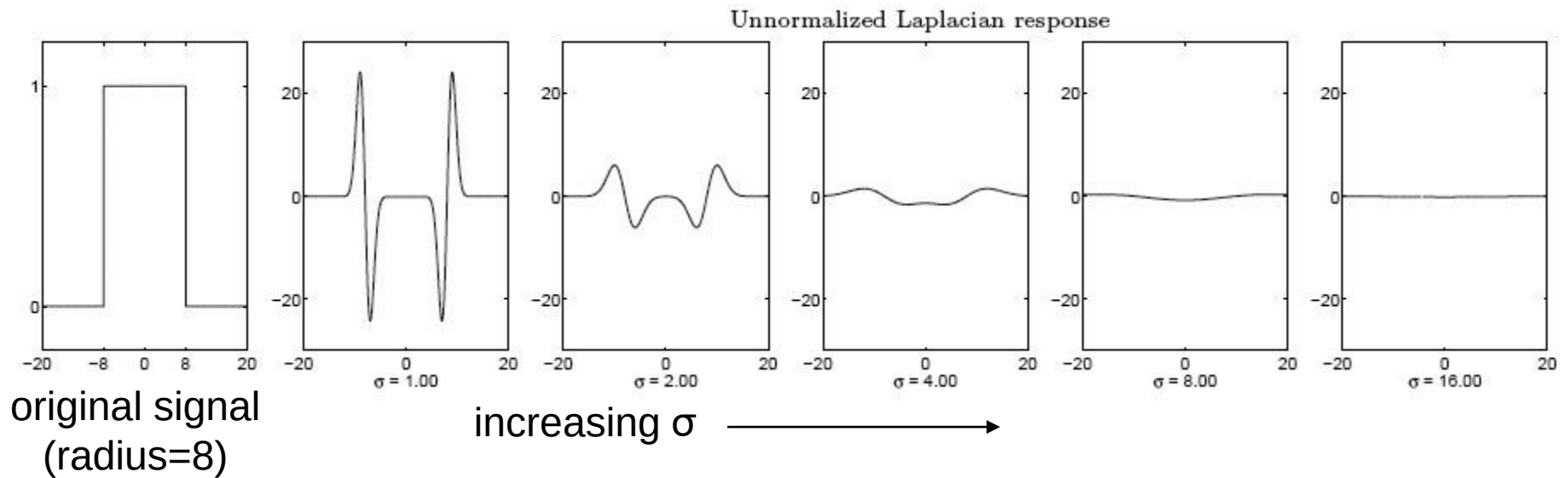
- Edge = ripple
- Blob = superposition of two ripples



Spatial selection: the magnitude of the Laplacian response will achieve a maximum at the center of the blob, provided the scale of the Laplacian is “matched” to the scale of the blob

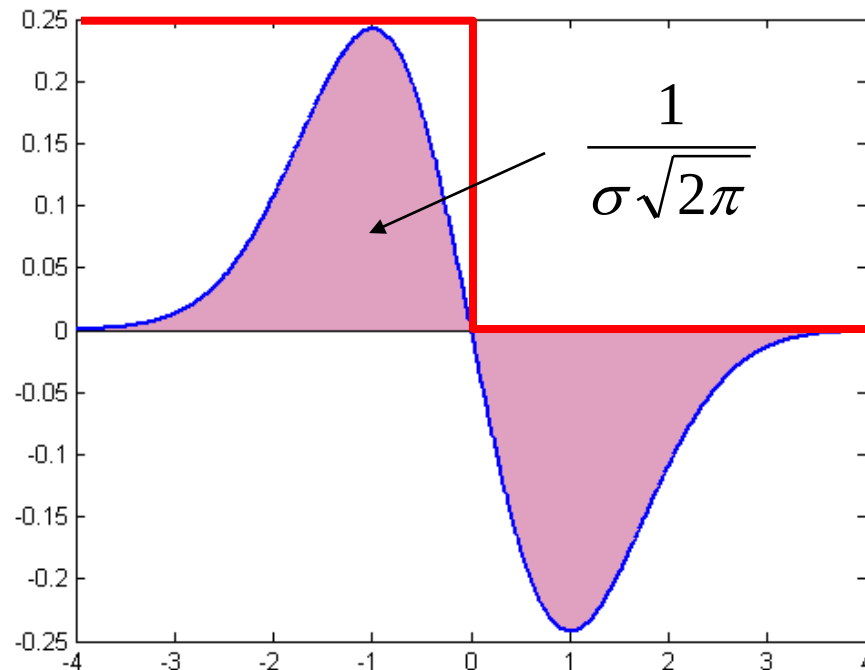
Scale selection

- We want to find the characteristic scale of the blob by convolving it with Laplacians at several scales and looking for the maximum response
- However, Laplacian response decays as scale increases:



Scale normalization

- The response of a derivative of Gaussian filter to a perfect step edge decreases as σ increases

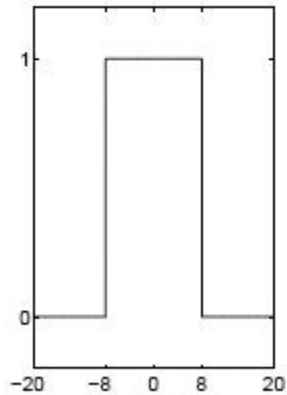


Scale normalization

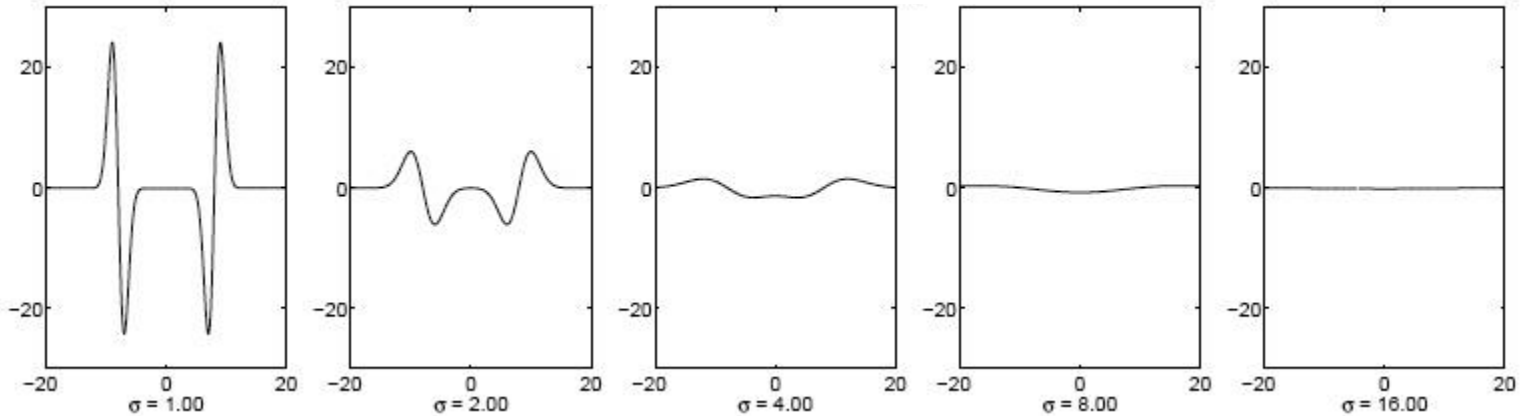
- The response of a derivative of Gaussian filter to a perfect step edge decreases as σ increases
- To keep response the same (scale-invariant), must multiply Gaussian derivative by σ
- Laplacian is the second Gaussian derivative, so it must be multiplied by σ^2

Effect of scale normalization

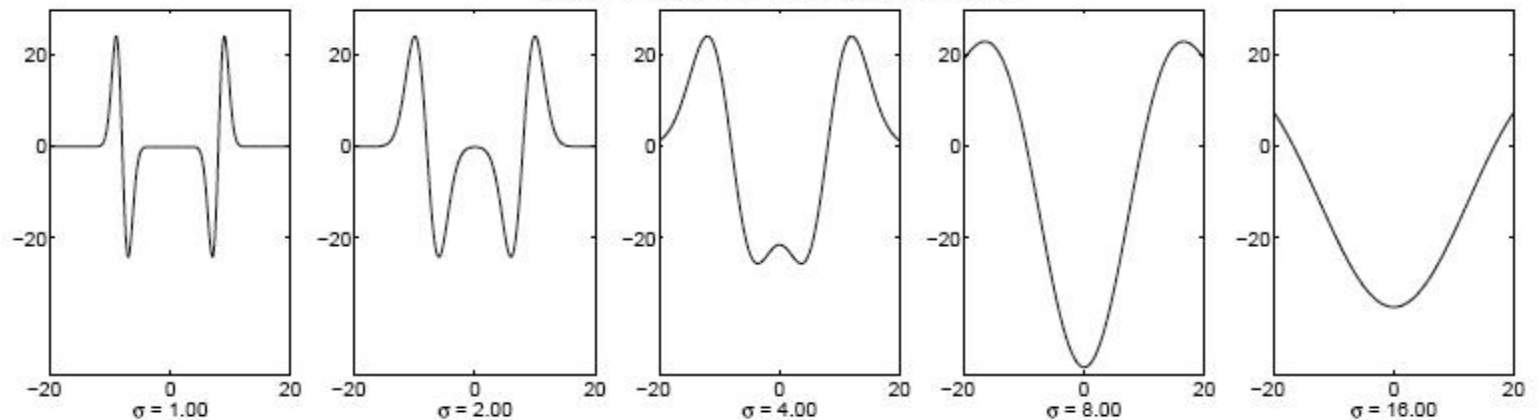
Original signal



Unnormalized Laplacian response



Scale-normalized Laplacian response

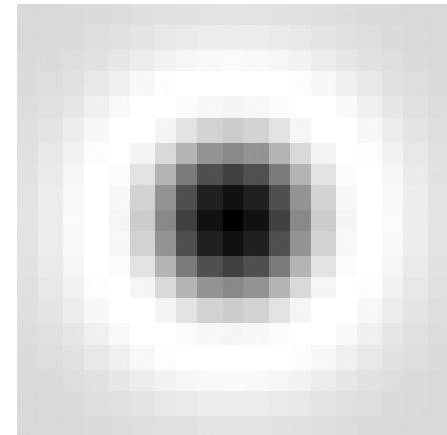
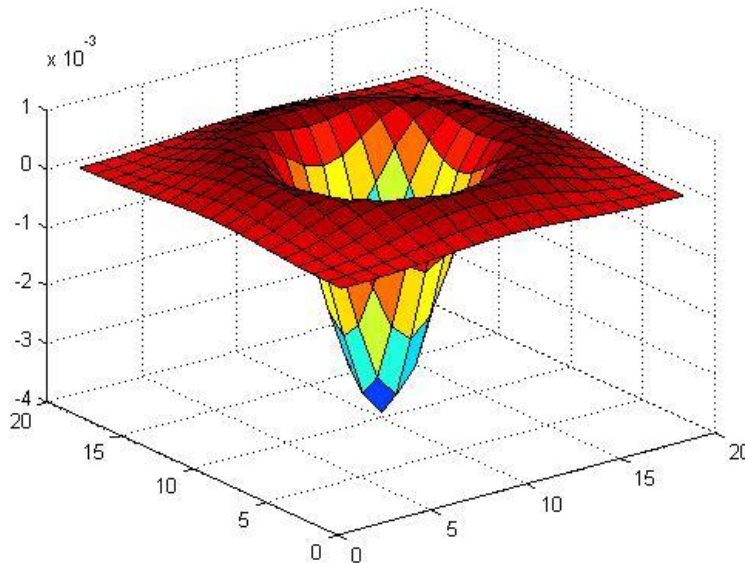


↑
maximum

Blob detection in 2D

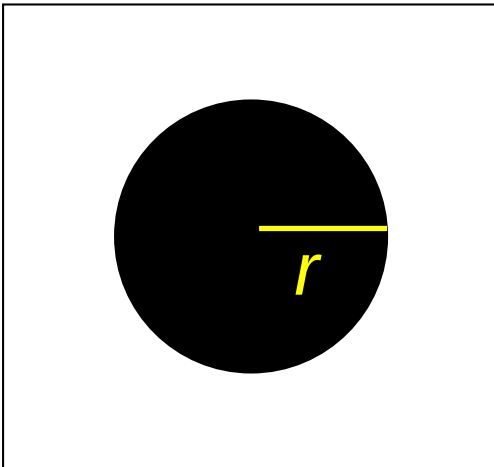
- *Scale-normalized* Laplacian of Gaussian:

$$\nabla_{\text{norm}}^2 g = \sigma^2 \left(\frac{\partial^2 g}{\partial x^2} + \frac{\partial^2 g}{\partial y^2} \right)$$

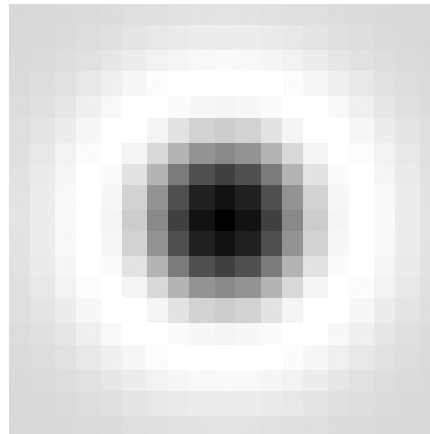


Blob detection in 2D

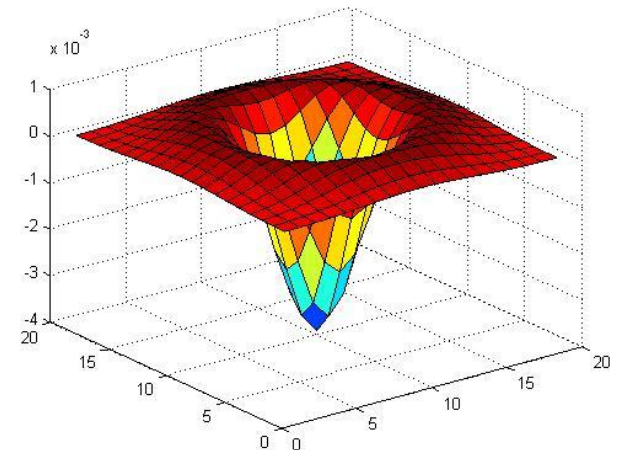
- At what scale does the Laplacian achieve a maximum response to a binary circle of radius r ?



image

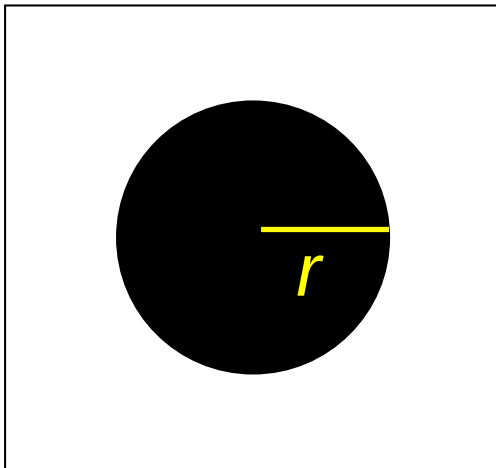


Laplacian

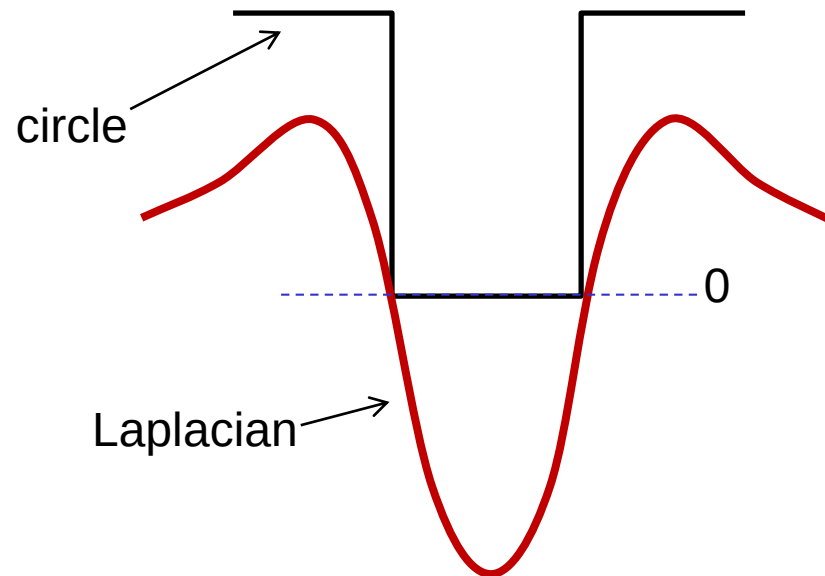


Blob detection in 2D

- At what scale does the Laplacian achieve a maximum response to a binary circle of radius r ?
- To get maximum response, the zeros of the Laplacian have to be aligned with the circle
- The Laplacian is given by (up to scale):
$$(x^2 + y^2 - 2\sigma^2) e^{-(x^2 + y^2)/2\sigma^2}$$
- Therefore, the maximum response occurs at $\sigma = r / \sqrt{2}$.



image



Scale-space blob detector

1. Convolve image with scale-normalized Laplacian at several scales

Scale-space blob detector: Example



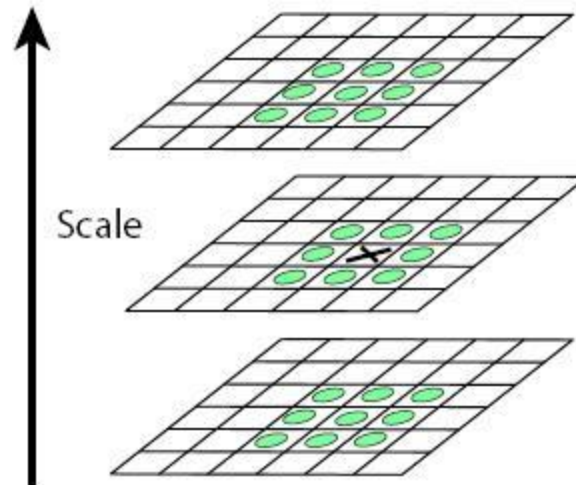
Scale-space blob detector: Example



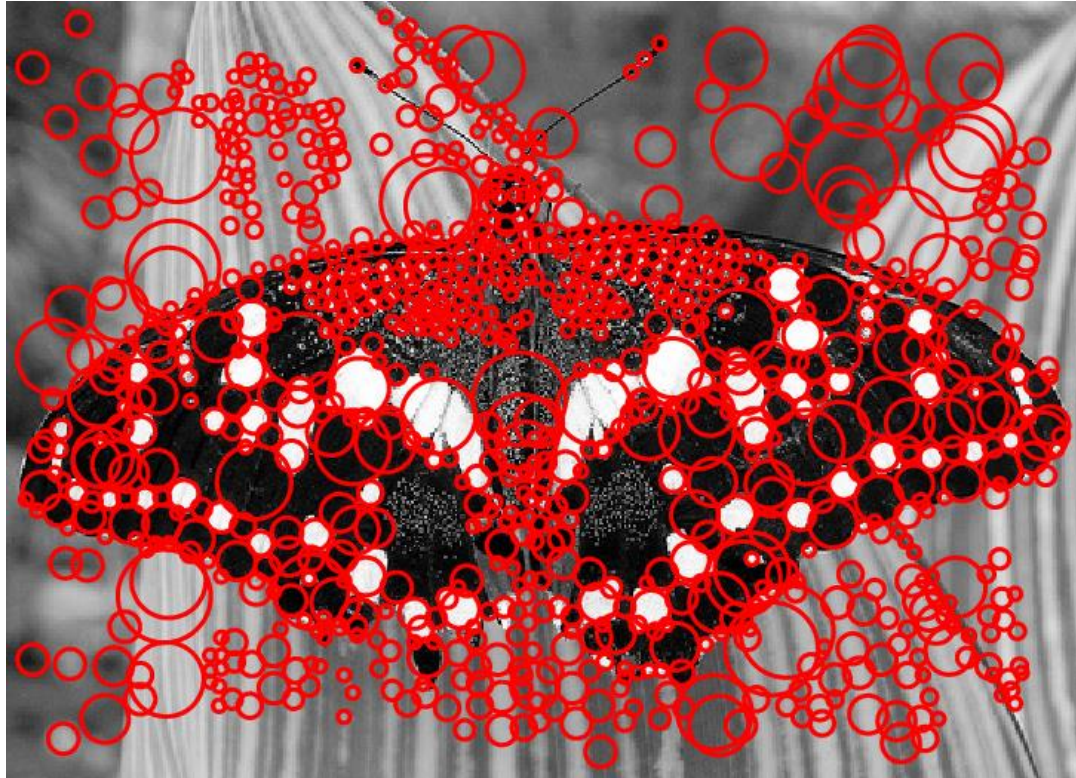
sigma = 11.9912

Scale-space blob detector

1. Convolve image with scale-normalized Laplacian at several scales
2. Find maxima of squared Laplacian response in scale-space



Scale-space blob detector: Example



Efficient implementation

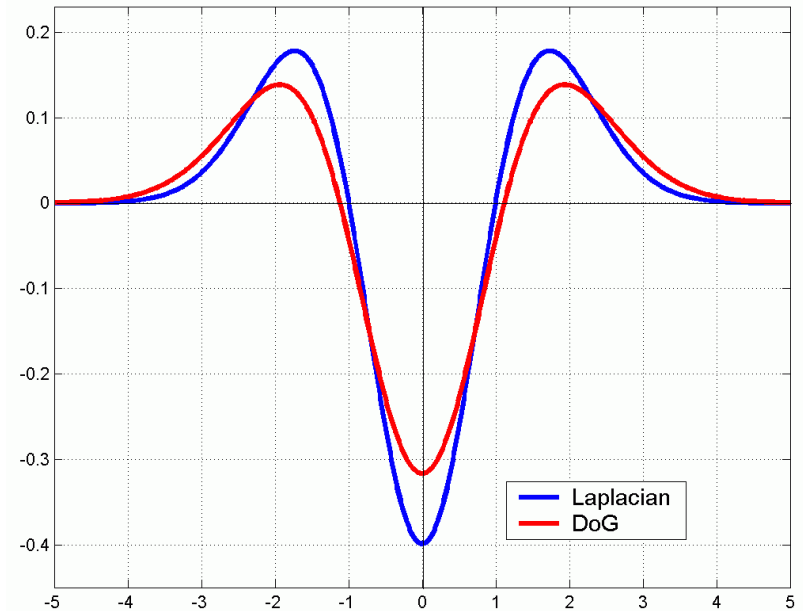
- Approximating the Laplacian with a difference of Gaussians:

$$L = \sigma^2 \left(G_{xx}(x, y, \sigma) + G_{yy}(x, y, \sigma) \right)$$

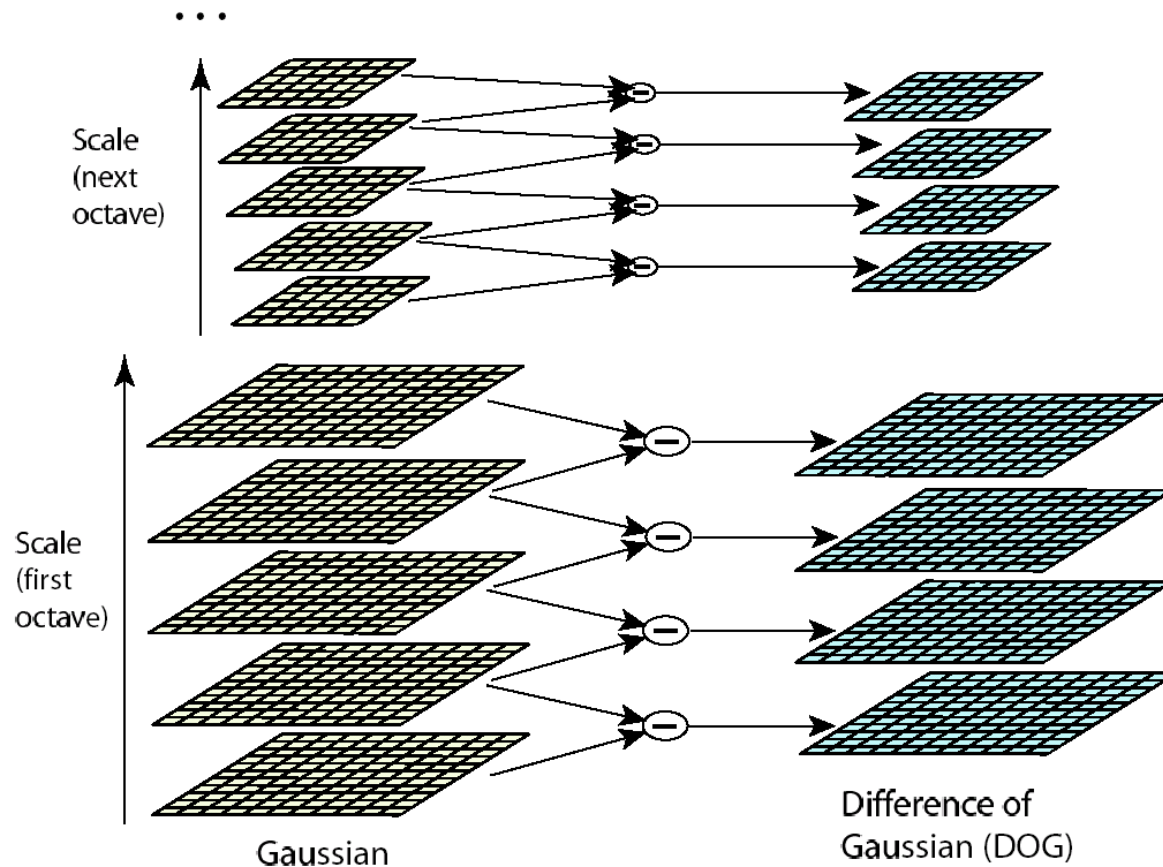
(Laplacian)

$$DoG = G(x, y, k\sigma) - G(x, y, \sigma)$$

(Difference of Gaussians)



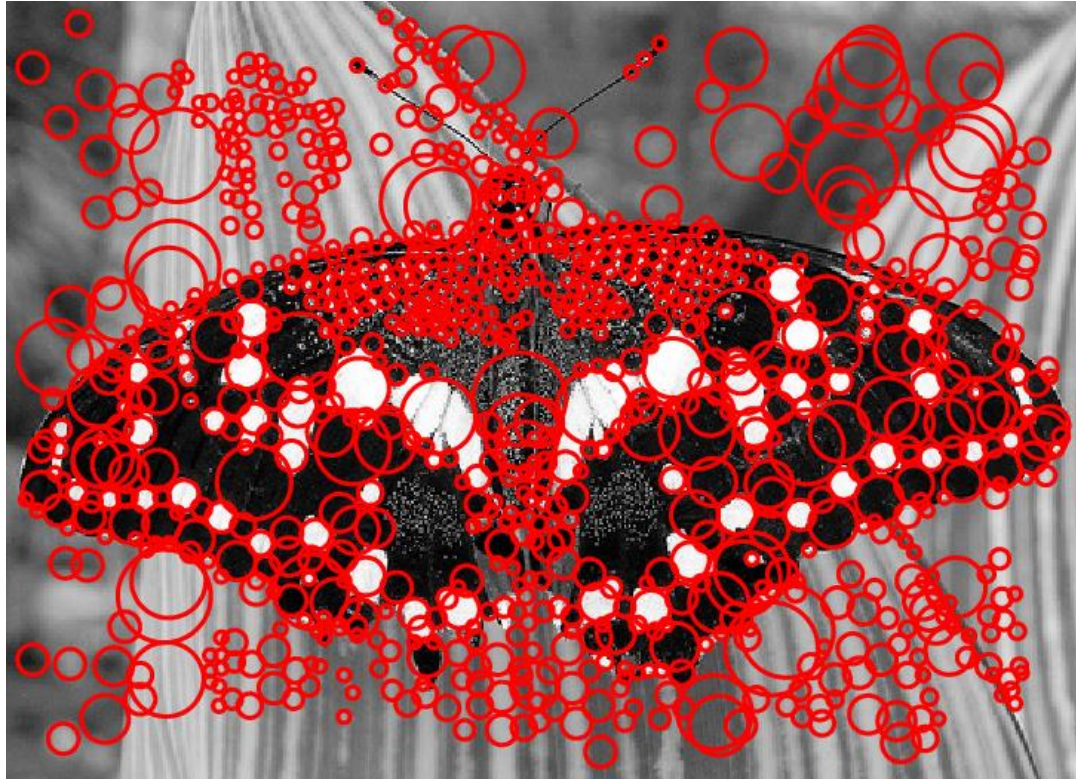
Efficient implementation



David G. Lowe. ["Distinctive image features from scale-invariant keypoints."](#) *IJCV* 60 (2), pp. 91-110, 2004.

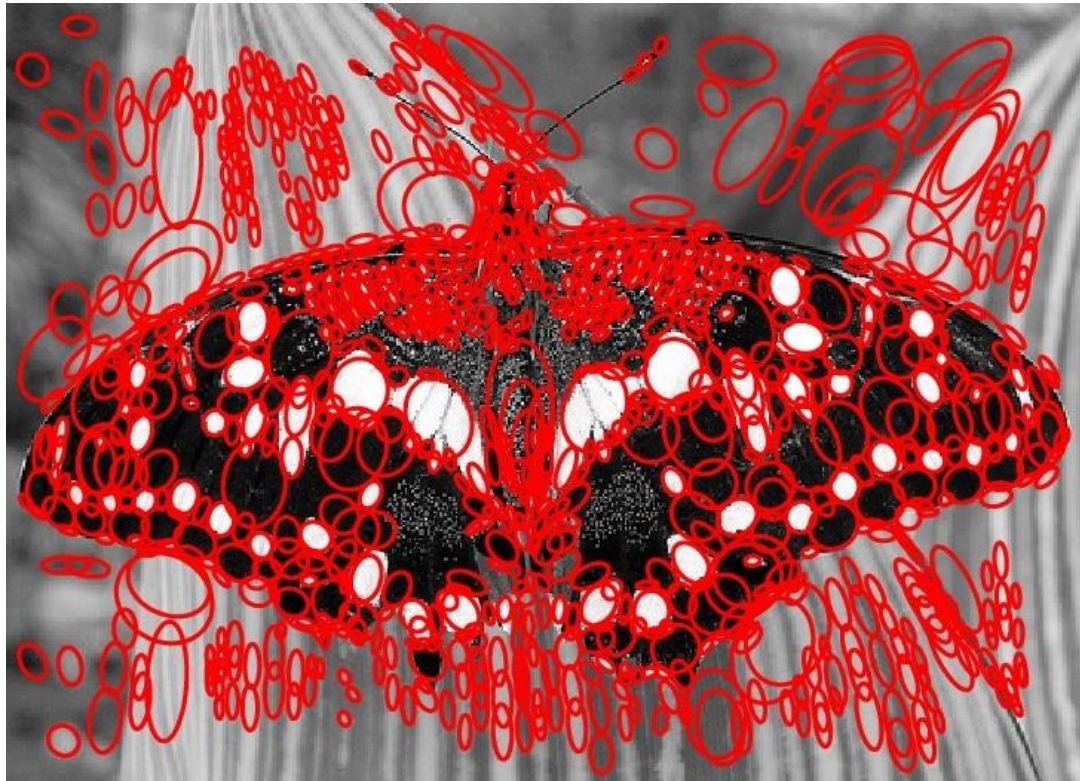
Eliminating edge responses

- Laplacian has strong response along edge



Eliminating edge responses

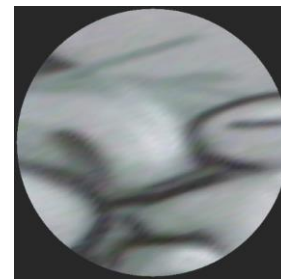
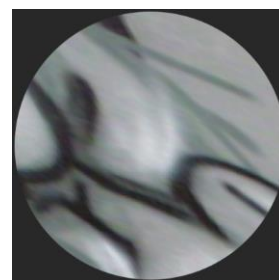
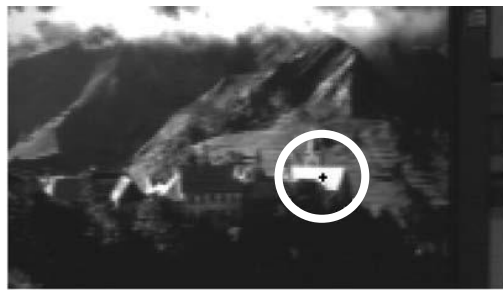
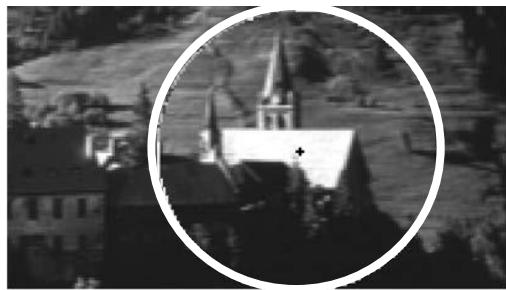
- Laplacian has strong response along edge



- Solution: filter based on Harris response function over neighborhoods containing the “blobs”

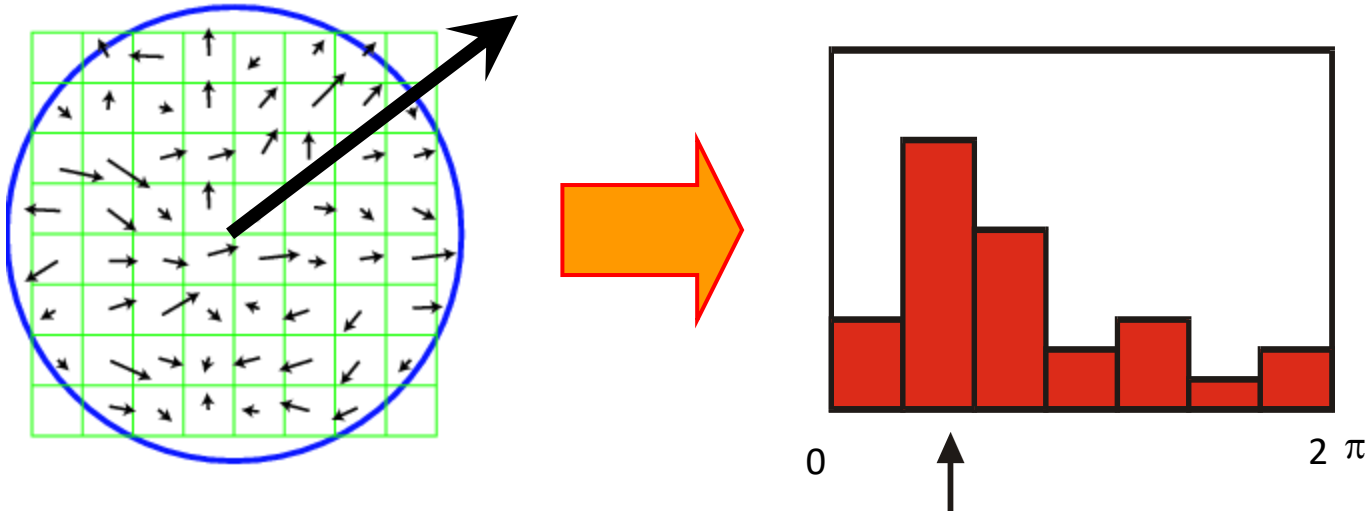
From feature detection to feature description

- Scaled and rotated versions of the same neighborhood will give rise to blobs that are related by the same transformation
- What to do if we want to compare the appearance of these image regions?
 - *Normalization*: transform these regions into same-size circles rotated to a consistent reference orientation



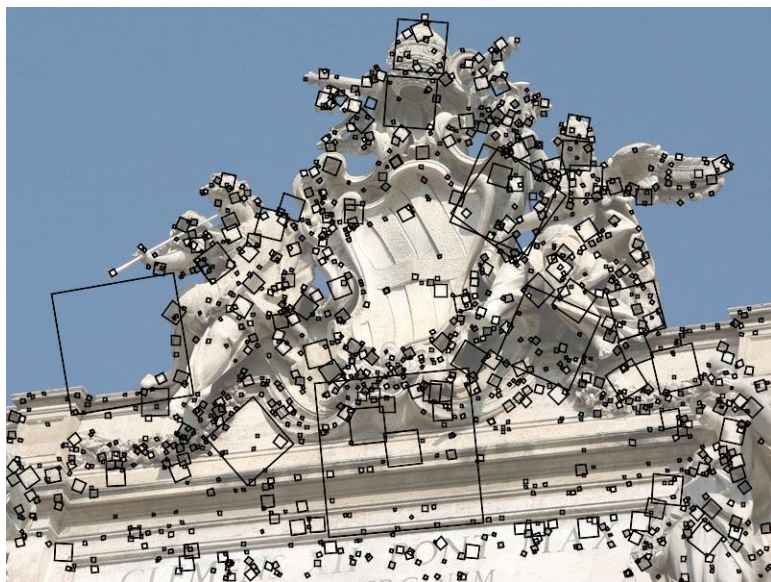
Finding a reference orientation

- Create histogram of local gradient directions in the patch
- Assign reference orientation at peak of smoothed histogram



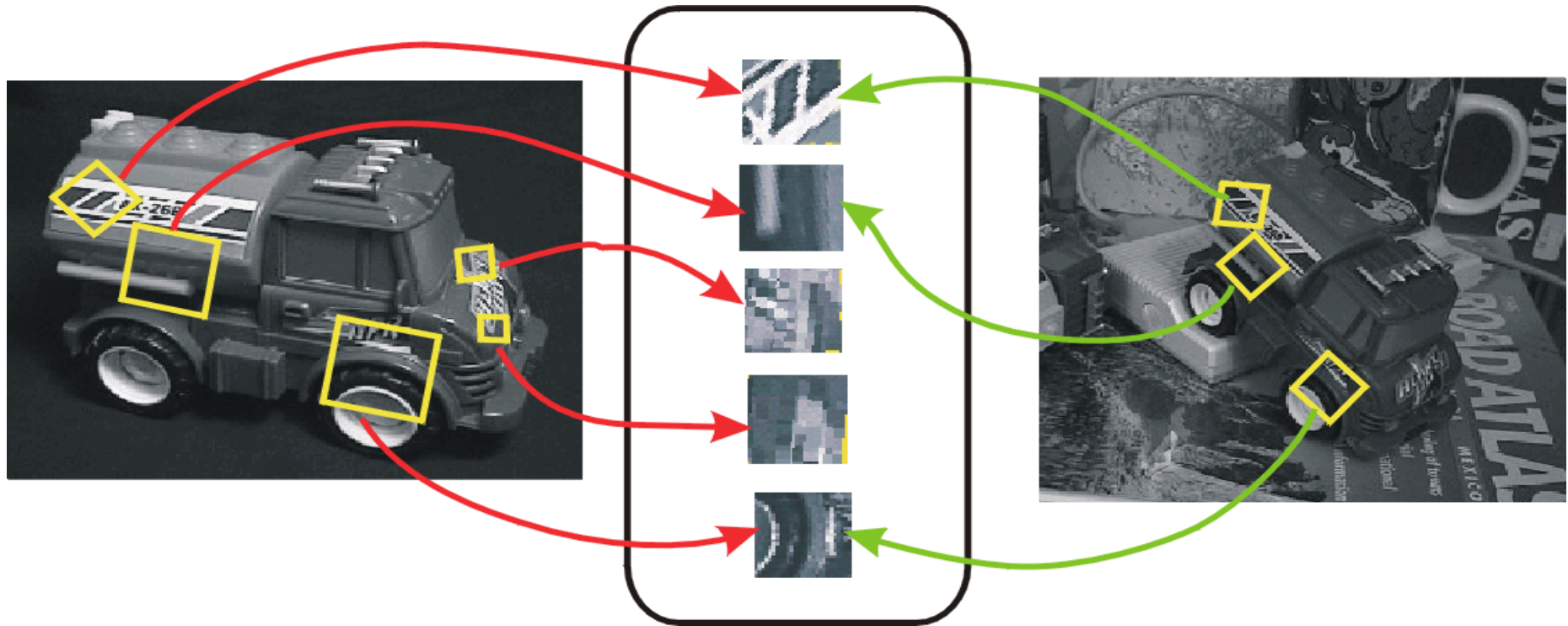
SIFT features

- Detected features with characteristic scales and orientations:



David G. Lowe. ["Distinctive image features from scale-invariant keypoints."](#) *IJCV* 60 (2), pp. 91-110, 2004.

From keypoint detection to feature description



Detection is *covariant*:

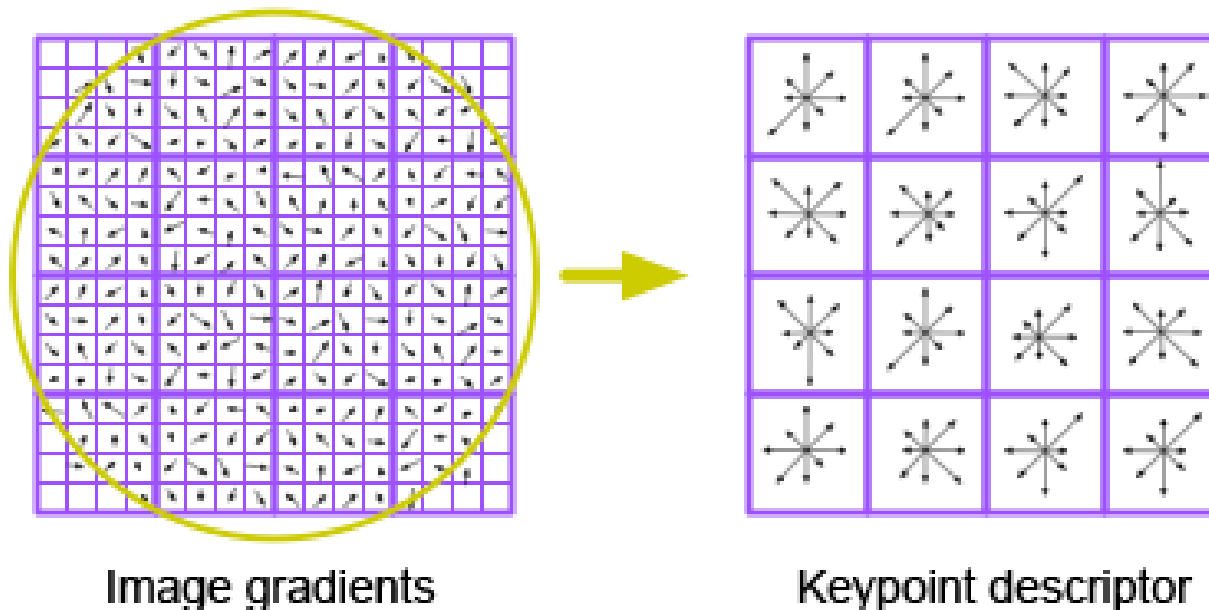
$$\text{features}(\text{transform}(\text{image})) = \text{transform}(\text{features}(\text{image}))$$

Description is *invariant*:

$$\text{features}(\text{transform}(\text{image})) = \text{features}(\text{image})$$

SIFT descriptors

- Inspiration: complex neurons in the primary visual cortex



D. Lowe. [Distinctive image features from scale-invariant keypoints.](#)
IJCV 60 (2), pp. 91-110, 2004.

Properties of SIFT

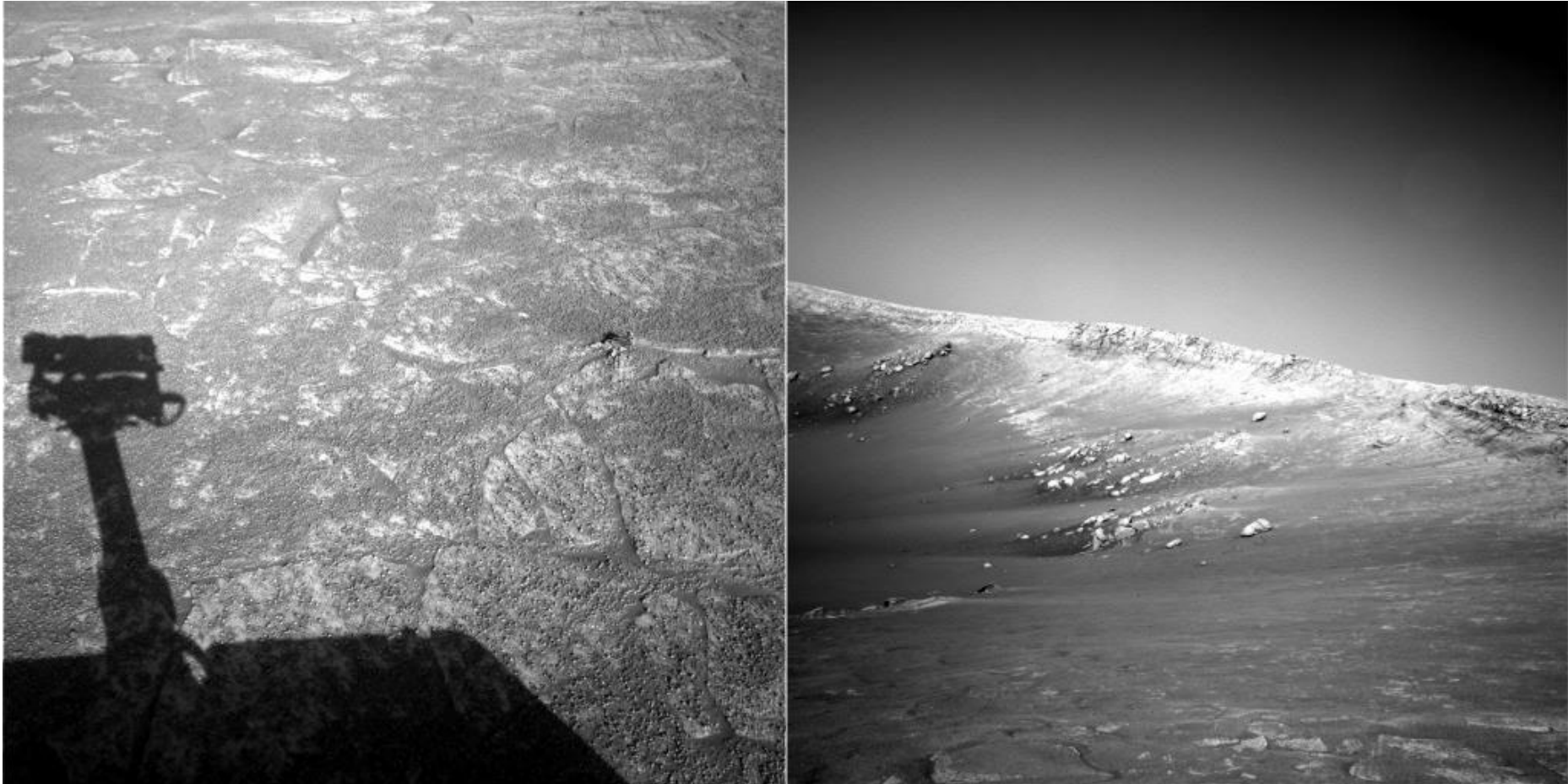
Extraordinarily robust detection and description technique

- Can handle changes in viewpoint
 - Up to about 60 degree out-of-plane rotation
- Can handle significant changes in illumination
 - Sometimes even day vs. night
- Fast and efficient—can run in real time
- Lots of code available



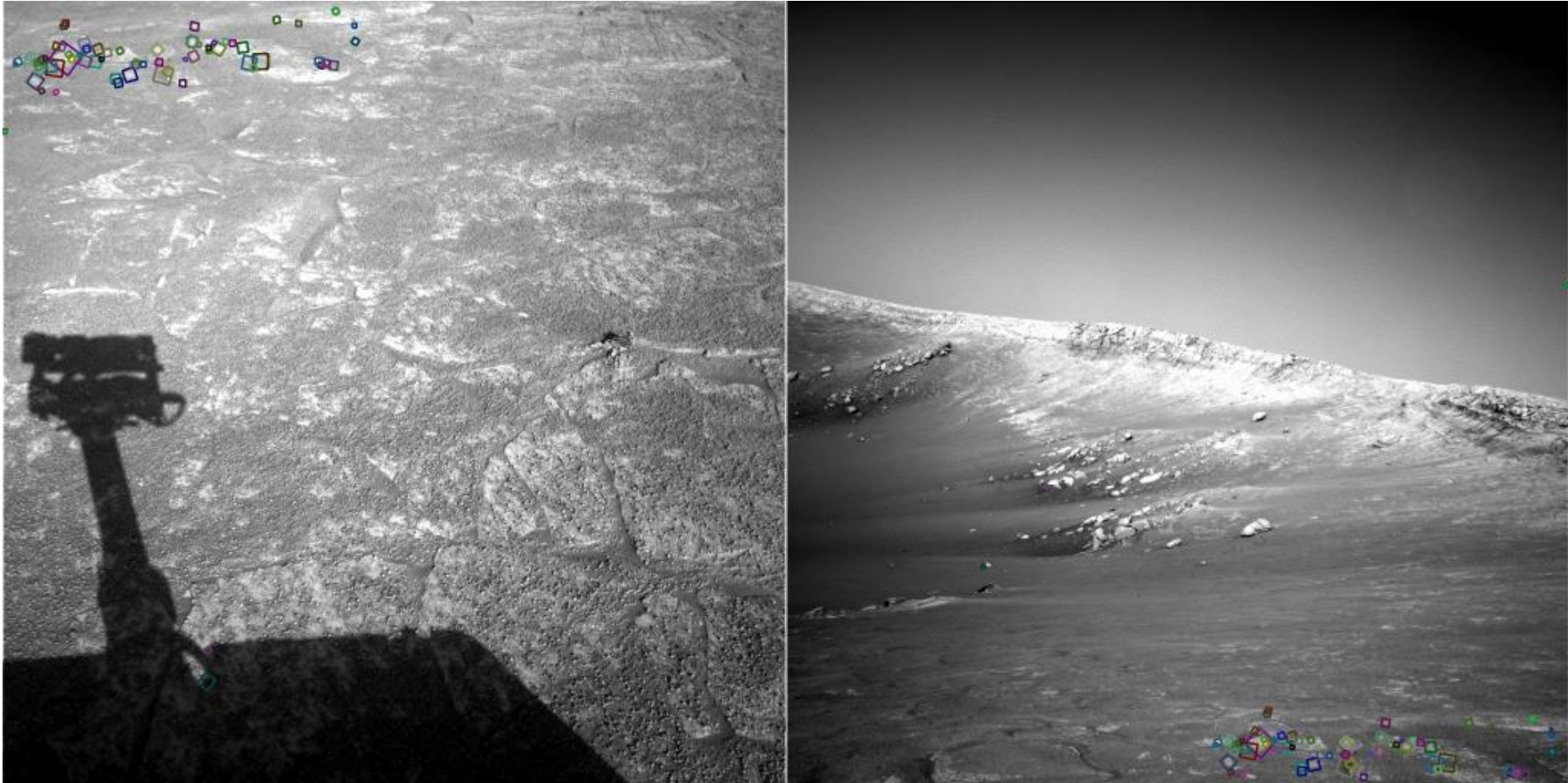
Source: N. Snavely

A hard keypoint matching problem



NASA Mars Rover images

Answer below (look for tiny colored squares...)



NASA Mars Rover images
with SIFT feature matches
Figure by Noah Snavely

What about 3D rotations?



What about 3D rotations?

- Affine transformation approximates viewpoint changes for roughly planar objects and roughly orthographic cameras



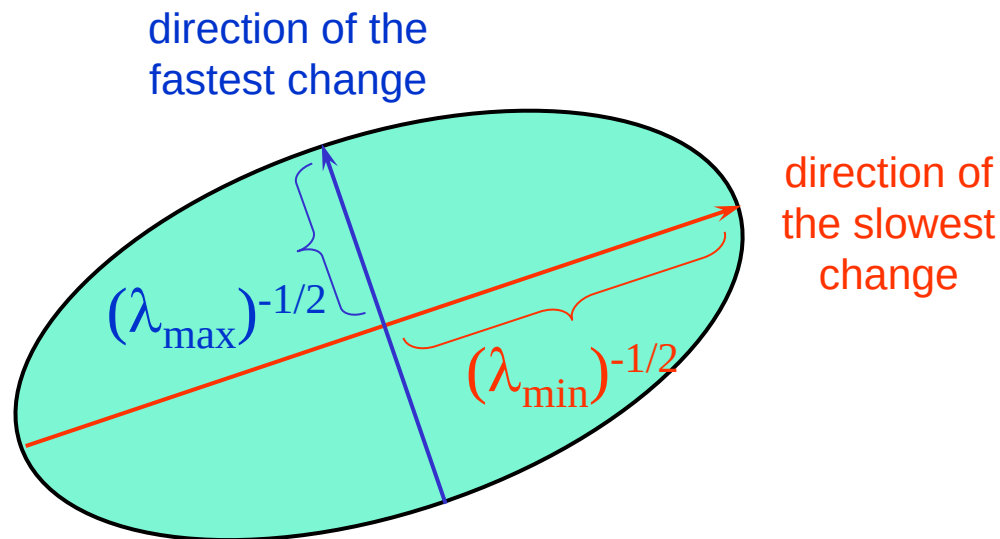
Affine adaptation

Consider the second moment matrix of the window containing the blob:

$$M = \sum_{x,y} w(x,y) \begin{bmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{bmatrix} = R^{-1} \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} R$$

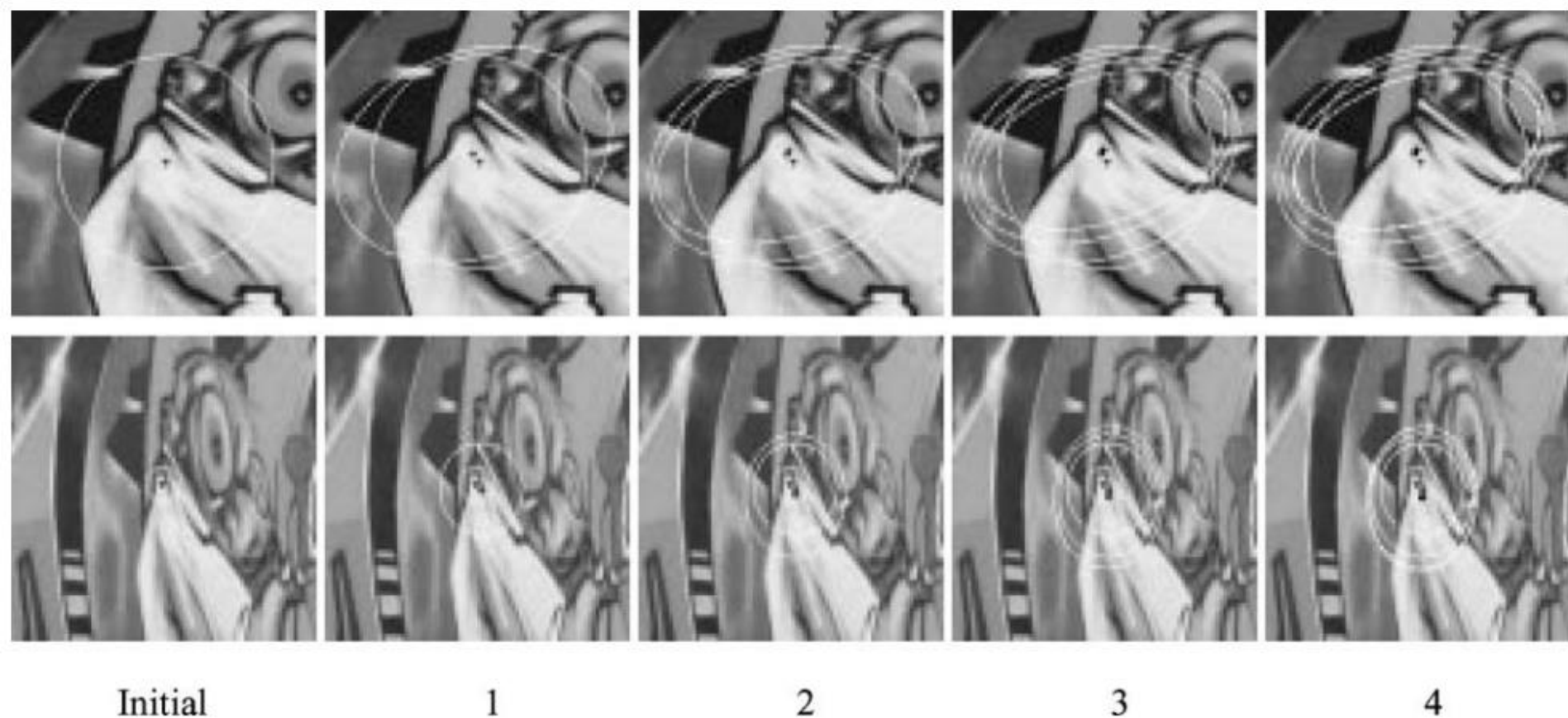
Recall:

$$\begin{bmatrix} u & v \end{bmatrix} M \begin{bmatrix} u \\ v \end{bmatrix} = \text{const}$$



This ellipse visualizes the “characteristic shape” of the window

Affine adaptation



K. Mikolajczyk and C. Schmid. [Scale and affine invariant interest point detectors](#). IJCV 60(1):63-86, 2004.

Keypoint detectors/descriptors for recognition: A retrospective

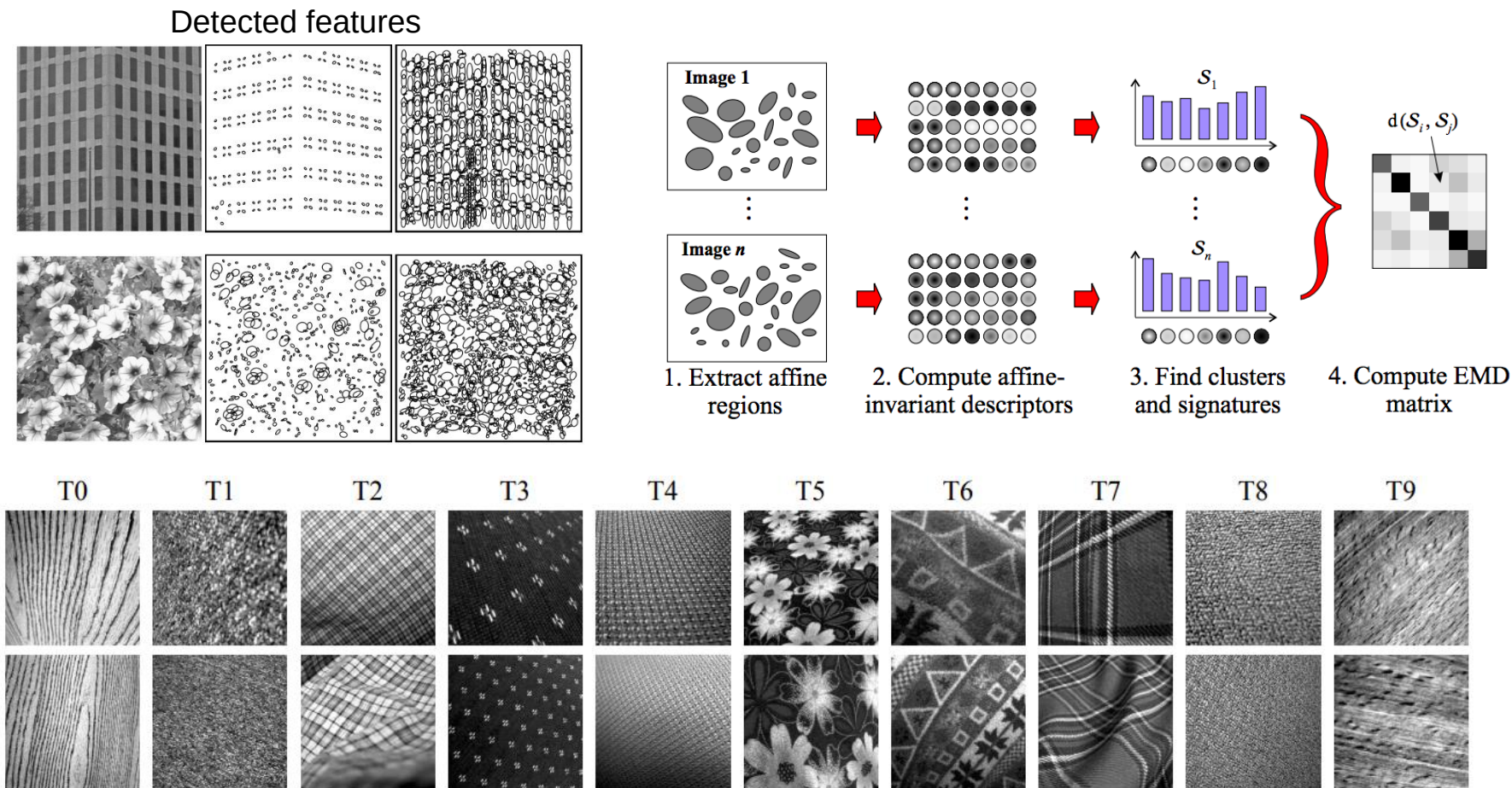
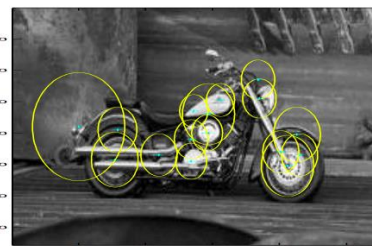


Figure 5: Samples of the ten texture classes used in the experiments of Section 3.2.

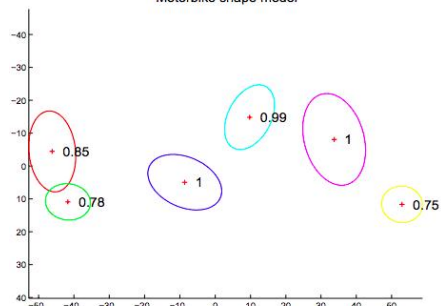
S. Lazebnik, C. Schmid, and J. Ponce, [A Sparse Texture Representation Using Affine-Invariant Regions](#), CVPR 2003

Keypoint detectors/descriptors for recognition: A retrospective

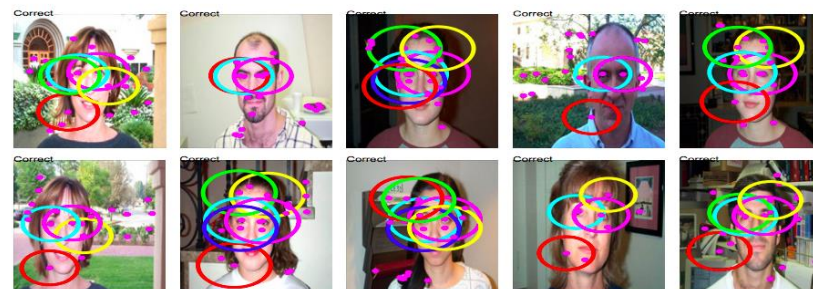
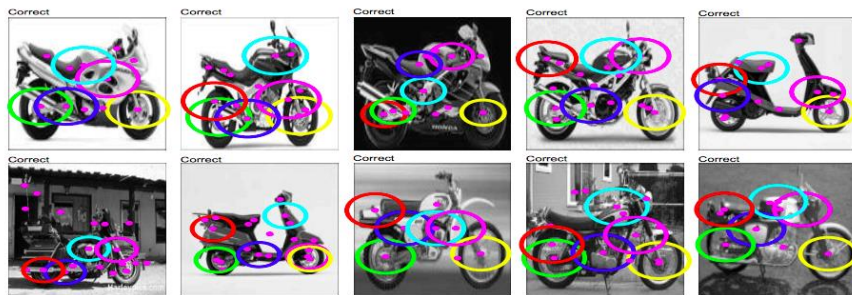
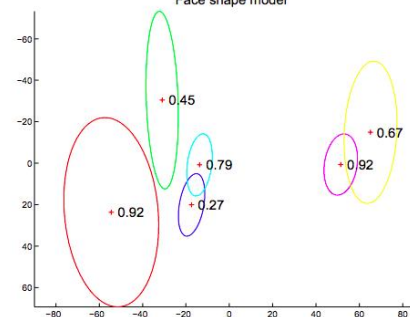
Detected features



Motorbike shape model

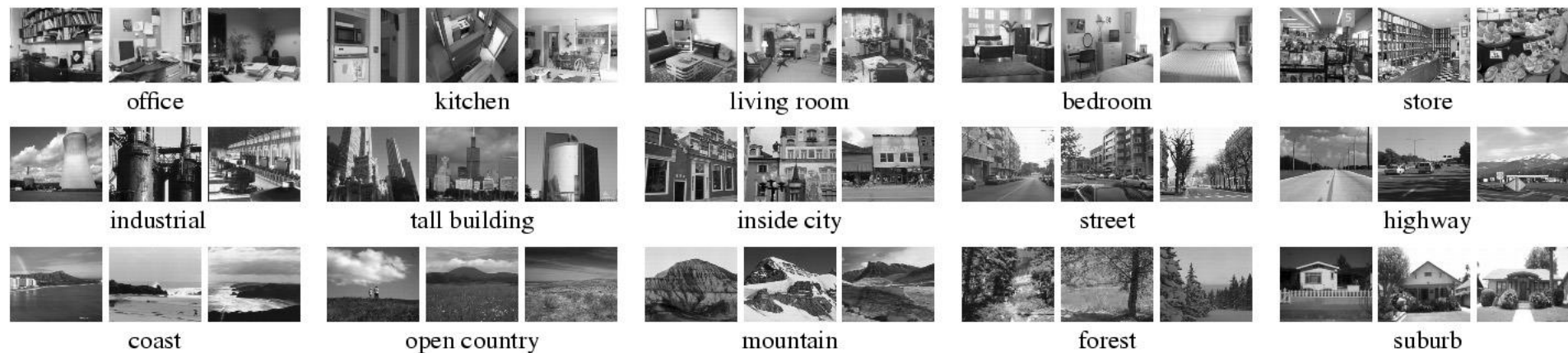


Face shape model



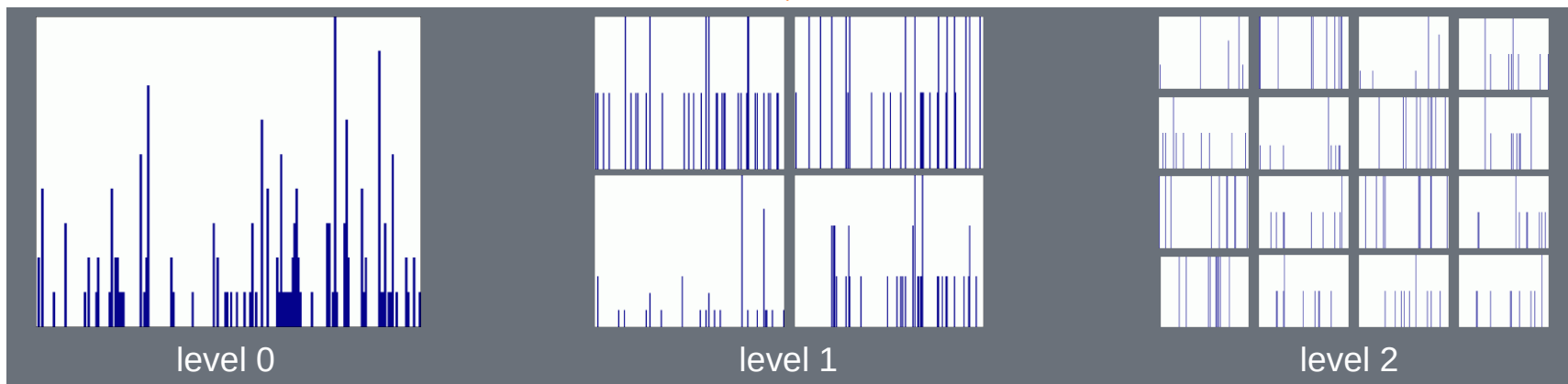
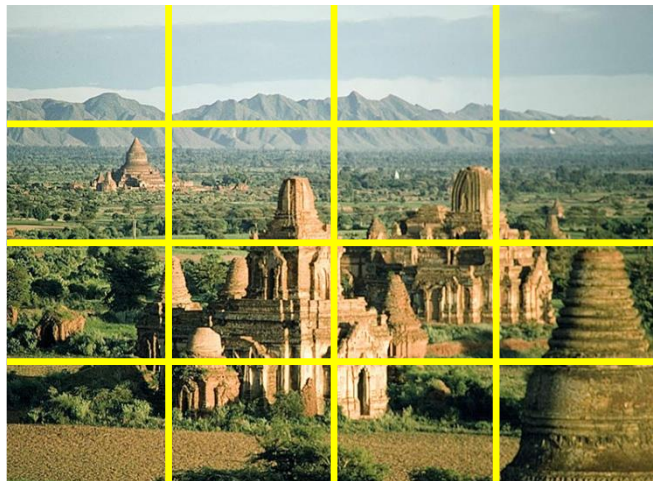
R. Fergus, P. Perona, and A. Zisserman, [Object Class Recognition by Unsupervised Scale-Invariant Learning](#), CVPR 2003 – winner of 2013 Longuet-Higgins Prize

Keypoint detectors/descriptors for recognition: A retrospective



S. Lazebnik, C. Schmid, and J. Ponce, [Beyond Bags of Features: Spatial Pyramid Matching for Recognizing Natural Scene Categories](#), CVPR 2006 – *winner of 2016 Longuet-Higgins Prize*

Keypoint detectors/descriptors for recognition: A retrospective



S. Lazebnik, C. Schmid, and J. Ponce, [Beyond Bags of Features: Spatial Pyramid Matching for Recognizing Natural Scene Categories](#), CVPR 2006 – winner of 2016 Longuet-Higgins Prize

Keypoint detectors/descriptors for recognition: A retrospective

 **Yann LeCun**
February 9 at 12:30pm · 🌐

Amusing how some computer vision researchers jokingly refer to work done before 2012 as "prehistoric".

 Like  Comment  Share

   Alex Berg, Bryan Russell and 693 others

29 Shares

 **Serge Belongie** I call it "antedeepluvian" (before the flood of deep learning papers)
   46
Like · Reply · 5d
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 **Yann LeCun** Yeah, I like this porte manteau.  3
Like · Reply · 4d

 **Alex Berg** "some"... 😊
Like · Reply · 4d

 **Yann LeCun** Yes, some. And jokingly.  1
Like · Reply · 4d

 **Aaron Hertzmann** I'm just thrilled when I get a paper to review that even acknowledges the existence of related work published prior to 2012.
Like · Reply · 4d  9

 **Alyosha Efros** Around 2005, I was at a vision workshop in MSRI where one of the Gemans said: "physics before Newton is now called 'miscellaneous early efforts'. Vision is at this same stage." Jury still out if we've seen our Newton yet. But looking hopeful.
Like · Reply · 5d · Edited  9